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import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("customer_support_tickets 3.csv")

print("="*60)
print("DATASET OVERVIEW")
print("="*60)

print("\nShape:", df.shape)
print("\nColumns:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

print("\nSummary Statistics:")
print(df.describe(include='all'))

status = df['Ticket Status'].value_counts()

plt.figure(figsize=(8,5))
status.plot(kind='bar')
plt.title("Ticket Status Distribution")
plt.xlabel("Status")
plt.ylabel("Count")
plt.show()

priority = df['Ticket Priority'].value_counts()

plt.figure(figsize=(7,7))
priority.plot(kind='pie', autopct='%1.1f%%')
plt.title("Ticket Priority Distribution")
plt.ylabel("")
plt.show()

ticket_type = df['Ticket Type'].value_counts()

plt.figure(figsize=(10,5))
ticket_type.plot(kind='barh')
plt.title("Ticket Type Analysis")
plt.xlabel("Count")
plt.show()

channel = df['Ticket Channel'].value_counts()

plt.figure(figsize=(8,5))
channel.plot(kind='bar')
plt.title("Tickets by Channel")
plt.xlabel("Channel")
plt.ylabel("Count")
plt.show()

gender = df['Customer Gender'].value_counts()

plt.figure(figsize=(7,5))
gender.plot(kind='bar')
plt.title("Customer Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()

products = df['Product Purchased'].value_counts().head(10)

plt.figure(figsize=(12,5))
products.plot(kind='bar')
plt.title("Top 10 Products with Most Tickets")
plt.xlabel("Product")
plt.ylabel("Tickets")
plt.xticks(rotation=45)

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plt.show()

plt.figure(figsize=(8,5))
df['Customer Age'].plot(kind='hist', bins=15)
plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.show()

rating = pd.to_numeric(df['Customer Satisfaction Rating'], errors='coerce')

plt.figure(figsize=(8,5))
rating.dropna().plot(kind='hist', bins=5)
plt.title("Customer Satisfaction Rating")
plt.xlabel("Rating")
plt.show()

plt.figure(figsize=(10,6))
pd.crosstab(
    df['Ticket Status'],
    df['Ticket Priority']
).plot(kind='bar', stacked=True)

plt.title("Ticket Status vs Priority")
plt.ylabel("Count")
plt.show()

print("\n"+"="*60)
print("BUSINESS INSIGHTS")
print("="*60)

print("Total Tickets :", len(df))
print("Duplicate Records :", df.duplicated().sum())
print("Most Common Ticket Status :", status.idxmax())
print("Most Common Ticket Type :", ticket_type.idxmax())
print("Most Common Priority :", priority.idxmax())
print("Most Used Channel :", channel.idxmax())
print("Top Product :", products.idxmax())
print("Average Customer Rating :", round(rating.mean(),2))

summary = pd.DataFrame({
    "Metric": [
        "Total Records",
        "Total Columns",
        "Duplicate Records",
        "Average Rating"
    ],
    "Value": [
        len(df),
        len(df.columns),
        df.duplicated().sum(),
        round(rating.mean(),2)
    ]
})

insights = pd.DataFrame({
    "Business Insights": [
        f"Total tickets analysed: {len(df)}",
        f"Most common ticket status: {status.idxmax()}",
        f"Most common ticket type: {ticket_type.idxmax()}",
        f"Most common priority: {priority.idxmax()}",
        f"Most used support channel: {channel.idxmax()}",
        f"Top product: {products.idxmax()}",
        f"Average customer satisfaction: {round(rating.mean(),2)}/5",
        "No duplicate records found."
    ]
})

with pd.ExcelWriter("Dataset_Analysis_Report.xlsx", engine="openpyxl") as writer:

    summary.to_excel(writer, sheet_name="Summary", index=False)

    df.isnull().sum().reset_index().rename(
        columns={"index": "Column", 0: "Missing Values"}
    ).to_excel(writer, sheet_name="Missing Values", index=False)

    status.reset_index().rename(
        columns={"index": "Ticket Status", "Ticket Status": "Count"}

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).to_excel(writer, sheet_name="Ticket Status", index=False)

priority.reset_index().rename(
    columns={"index": "Priority", "Ticket Priority": "Count"}
).to_excel(writer, sheet_name="Priority", index=False)

channel.reset_index().rename(
    columns={"index": "Channel", "Ticket Channel": "Count"}
).to_excel(writer, sheet_name="Channels", index=False)

ticket_type.reset_index().rename(
    columns={"index": "Ticket Type", "Ticket Type": "Count"}
).to_excel(writer, sheet_name="Ticket Types", index=False)

gender.reset_index().rename(
    columns={"index": "Gender", "Customer Gender": "Count"}
).to_excel(writer, sheet_name="Gender", index=False)

products.reset_index().rename(
    columns={"index": "Product", "Product Purchased": "Count"}
).to_excel(writer, sheet_name="Top Products", index=False)

insights.to_excel(writer, sheet_name="Insights", index=False)

df.to_excel(writer, sheet_name="Raw Dataset", index=False)

print("\nDataset Analysis Completed Successfully!")
print("Excel File Saved As: Dataset_Analysis_Report.xlsx")
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DATASET OVERVIEW

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Shape: (8469, 17)

Columns:
'Ticket ID', 'Customer Name', 'Customer Email', 'Customer Age', 'Customer Gender', 'Product Purchased',

Data Types:
Ticket ID int64
Customer Name object
Customer Email object
Customer Age int64
Customer Gender object
Product Purchased object
Date of Purchase object
Ticket Type object
Ticket Subject object
Ticket Description object
Ticket Status object
Resolution object
Ticket Priority object
Ticket Channel object
First Response Time object
Time to Resolution object
Customer Satisfaction Rating float64
dtype: object

Missing Values:
Ticket ID 0
Customer Name 0
Customer Email 0
Customer Age 0
Customer Gender 0
Product Purchased 0
Date of Purchase 0
Ticket Type 0
Ticket Subject 0
Ticket Description 0
Ticket Status 0
Resolution 5700
Ticket Priority 0
Ticket Channel 0
First Response Time 2819
Time to Resolution 5700
Customer Satisfaction Rating 5700
dtype: int64

Duplicate Rows: 0

Summary Statistics:

	Ticket ID	Customer Name	Customer Email	Customer Age	\
count	8469.000000	8469	8469	8469.000000	
unique	NaN	8028	8320	NaN	
top	NaN	Michael Garcia	bsmith@example.com	NaN	
freq	NaN	5	4	NaN	
mean	4235.000000	NaN	NaN	44.026804	
std	2444.934048	NaN	NaN	15.296112	
min	1.000000	NaN	NaN	18.000000	
max	2118.000000	NaN	NaN	31.000000	
mean	4235.000000	NaN	NaN	44.000000	
std	6352.000000	NaN	NaN	57.000000	
min	8469.000000	NaN	NaN	70.000000	

	Customer Gender	Product Purchased	Date of Purchase	Ticket Type	\
count	8469	8469	8469	8469	
unique	3	42	730	5	
top	Male	Canon EOS	2020-10-21	Refund request	
freq	2896	240	24	1752	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

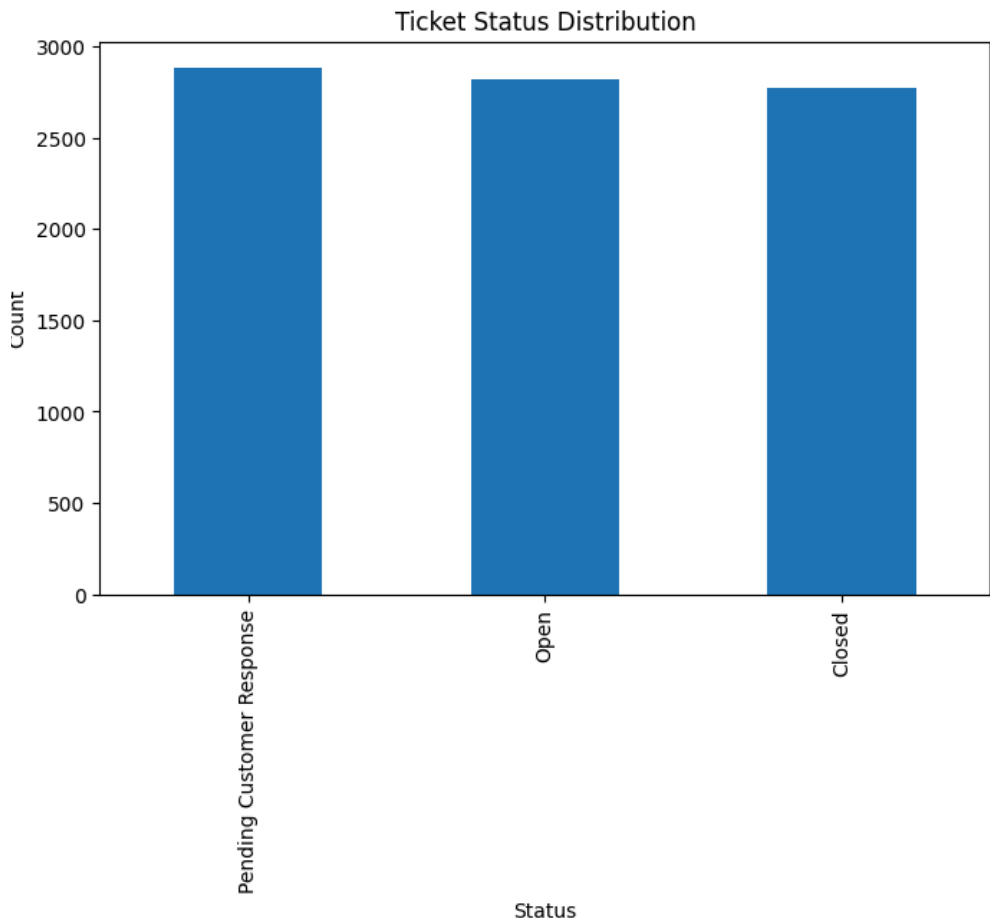
	Ticket Subject	Ticket Description	\
count	8469	8469	
unique	16	8077	
top	Refund request	I'm having an issue with the {product_purchase...	
freq	576	25	
mean	NaN	NaN	

id	NaN	NaN
in	NaN	NaN
%	NaN	NaN
%	NaN	NaN
%	NaN	NaN
ax	NaN	NaN

	Ticket Status	Resolution	Ticket Priority	\
ount	8469	2769	8469	
ique	3	2769	4	
op	Pending Customer Response	We seat culture plan.	Medium	
req	2881	1	2192	
ean	NaN	NaN	NaN	
id	NaN	NaN	NaN	
in	NaN	NaN	NaN	
%	NaN	NaN	NaN	
%	NaN	NaN	NaN	
%	NaN	NaN	NaN	
ax	NaN	NaN	NaN	

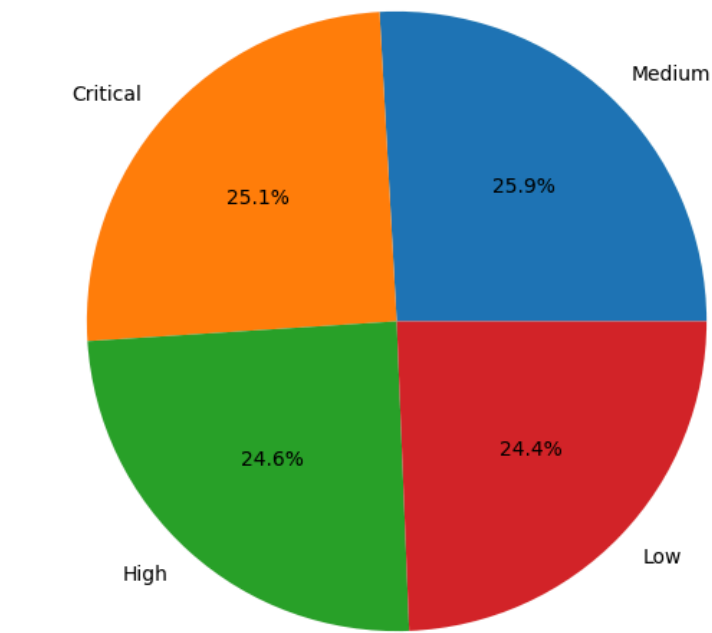
	Ticket Channel	First Response Time	Time to Resolution	\
ount	8469	5650	2769	
ique	4	5470	2728	
op	Email	2023-06-01 20:45:39	2023-06-01 17:14:42	
req	2143	3	3	
ean	NaN	NaN	NaN	
id	NaN	NaN	NaN	
in	NaN	NaN	NaN	
%	NaN	NaN	NaN	
%	NaN	NaN	NaN	
%	NaN	NaN	NaN	
ax	NaN	NaN	NaN	

	Customer Satisfaction Rating
ount	2769.000000
ique	NaN
op	NaN
req	NaN
ean	2.991333
id	1.407016
in	1.000000
%	2.000000
%	3.000000
%	4.000000
ax	5.000000

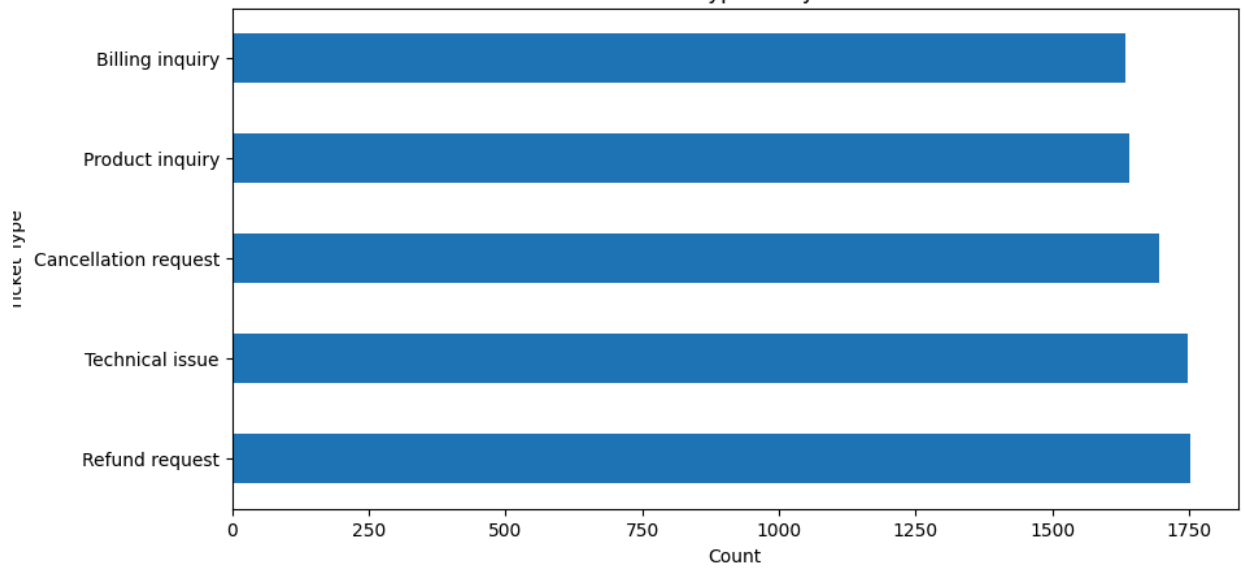


Ticket Priority Distribution

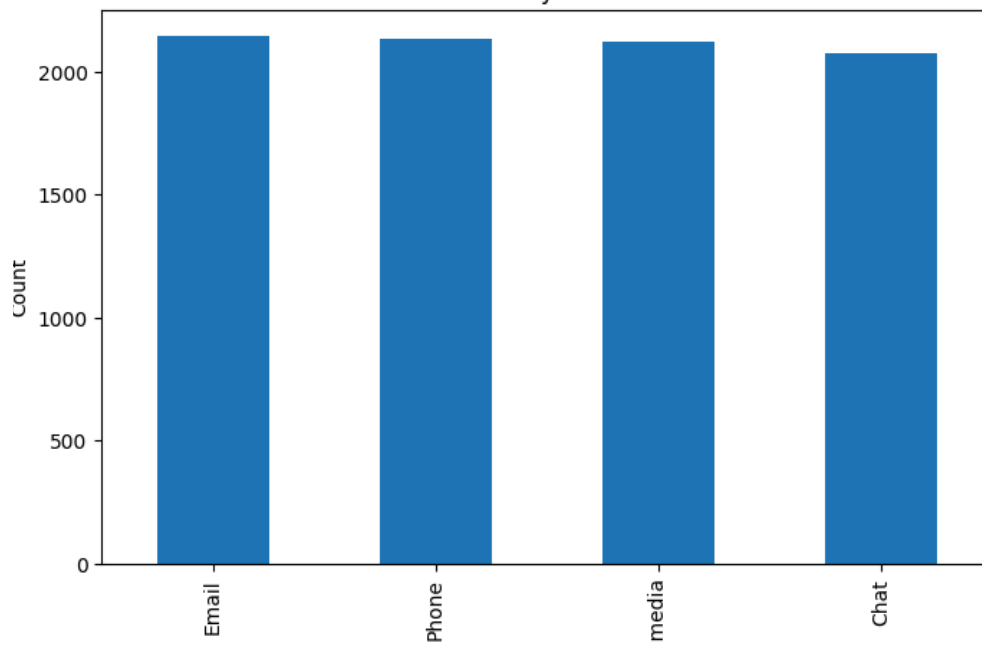
Ticket Priority Distribution

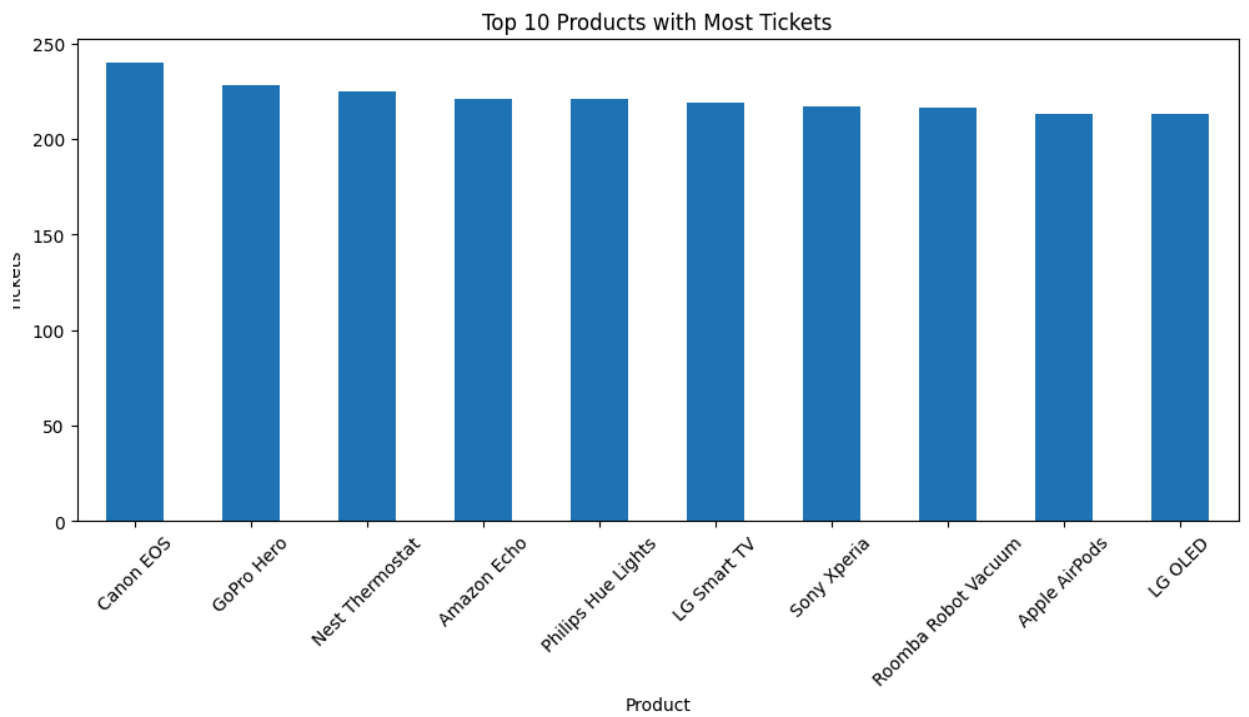
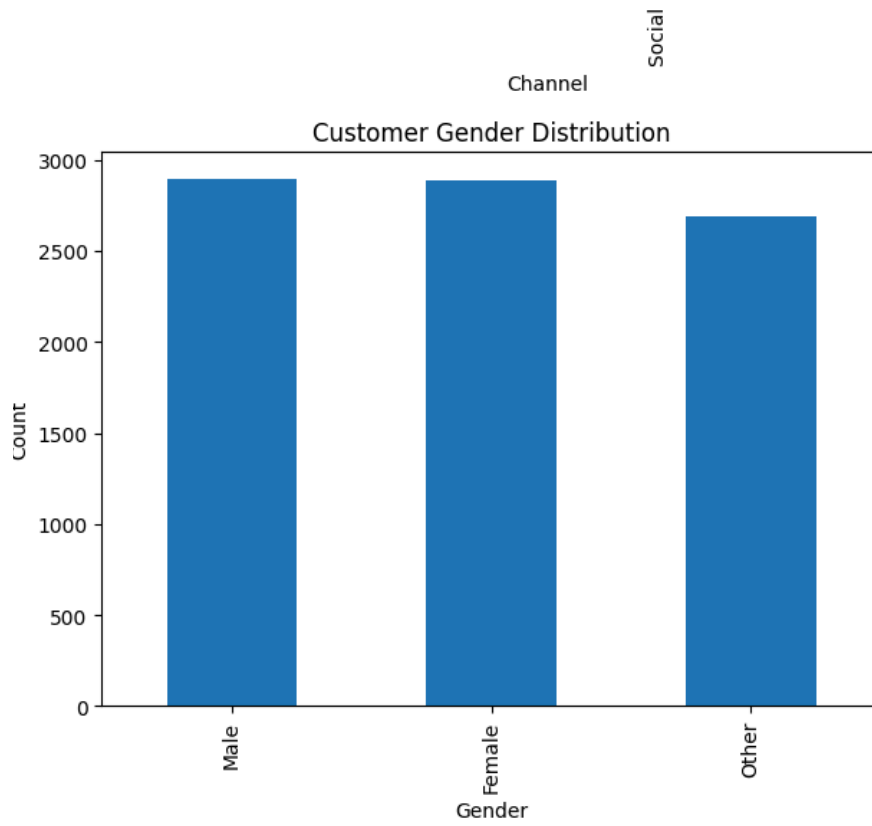


Ticket Type Analysis



Tickets by Channel





Customer Age Distribution

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